Program:	**Diploma in Integrated Circuit Design and Fabrication**		
Course Code:	**6371A**	Course Title:	**Computer Hardware and Networks**
Semester: **6**		Credits:	**4**
Course Category:	**Program Elective Course**		
Periods per week: **4 (L:4 T:0 P:0)**		Periods per semester:	**60**

Course Objectives:

- This course is designed to get a thorough knowledge of internal and external components of desktop computers.
- Provide adequate knowledge of the concept of data communication and computer networks
- Acquire knowledge of data sharing, transmission media and protocols.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics on Computer Fundamentals	1008	Introduction to IT systems lab.	1
Knowledge of electronics components	2031	Fundamentals of Electrical and Electronics Engineering	2
Number system, Binary arithmetic Error detection and correction	3044	Digital Electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Outline basic structure of a computer, motherboard components in desktop computers and BIOS settings	14	Understanding
CO2	Describe power supply and storage devices of a computer	14	Understanding
CO3	Describe the basic concepts of data communication and computer networks	15	Understanding

CO4	Outline services and protocols of network, transport layers and application layer	15	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Outline basic structure of a computer, motherboard components in desktop computers and BIOS settings**		
M1.01	Illustrate the functional units of a computer with block diagram.	3	Understanding
M1.02	Describe different connectors available in desktop computers	3	Understanding
M1.03	Explain various motherboard components and its features in desktop computer	4	Understanding
M1.04	Describe BIOS settings	4	Understanding
Contents: Functional units of a computer- Block diagram. **Desktop Motherboard components:** Motherboard form factors, Chipset, CPU Socket types, CPU fan & heatsink mounting points, Connectors for integrated peripherals, CMOS Backup battery, Bus & Interconnection slots, South bridge, North bridge, AGP slots, IDE & SATA Connectors- features, Memory slots, Switches & Jumpers. **BIOS:** Define booting & understand the concept of multi os booting, BIOS – settings & configuration			
CO2	**Describe power supply and storage devices of a computer**		
M2.01	Describe characteristics of Power Supplies.	3	Remembering

M2.02	Explain the details about powering the PC	5	Understanding
M2.03	List memory hierarchy	2	Understanding
M2.04	Discuss different types of primary memory and secondary memory devices	4	Understanding
	Series Test – I	1	

Contents:

Purpose and characteristics of Power Supplies- Power connectors- Replacement of Power Supplies, Overview of SMPS and its features.

UPS: Block diagram, Types of UPS.

Introduction to computer memory system, Memory characteristics & Hierarchy.

Different types of memory – Primary memory: RAM, ROM, Cache memory concepts & levels-Secondary memory – Direct access, Random access, Sequential access storage devices.

CO3	**Describe the basic concepts of data communication and computer networks**		
M3.01	Explain the basic concepts of Data communication	1	Understanding
M3.02	Compare types of networks and topology	2	Understanding
M3.03	Explain the layered concept of TCP/IP and OSI models	4	Understanding
M3.04	Describe the concept of physical layer	4	Understanding
M3.05	Outline the basic concepts of data link layer and services	4	Understanding

Contents:

Introduction to data communication – definition, components, data representations, data flow

Networks – definition, network criteria, types of connection, physical topology, network types, Internet.

TCP/IP and OSI models- functions of layers in both models, comparison, TCP/IP protocol suite.\

Physical layer –Transmission modes, Transmission media – Guided media, Unguided media.

Datalink layer –Datalink control- framing, list the types of framing, flow and error control.

Datalink layer protocols – For noiseless channel- simplest, stop -and- wait.

CO4	**Outline services and protocols of network, transport layers and application layer**		
M4.01	Summarize network layer services, IP addressing Control protocols and functions	4	Understanding

M4.02	Summarize transport layer services and protocols.	4	Understanding
M4.03	Describe the features and services of application layer.	4	Understanding
M4.04	Explain Application layer protocols	3	Understanding
	Series Test – II	1	

Contents:

Network layer- Network layer services.
Network layer protocols – Internet protocol, IP v4 protocol & IPv6 protocol- Packet Format.
Transport layer- Transport layer services
Transport layer protocols- UDP-User Datagram, Features & Services, Applications. TCP – Features & Services, Applications, Segment format.
Application layer - Application layer services. Application layer Paradigms- client-server, peer to peer
Application layer protocols – FTP & HTTP - Basic models.

Text/Reference

T/R	Book Title/Author
T1	Stephen J.Bigelow, Troubleshooting, Maintaining and Repairing PCs, TMH, Fifth Edition.
T2	Morris Rosenthal, An introduction to Troubleshooting and Repairing laptop.
T3	D Balasubramanian., Computer installation and servicing. TMH, 2010
R1	Behrouz A. Forouzan -Data Communications and Networking – McGraw Hill Edn. – Fourth Edition/Fifth Edition
R2	Complete A+ Guide to IT Hardware and Software: AA CompTIA A+ Core 1 (220-1001) & CompTIA A+ Core 2 (220-1002)
R3	Joel Rosenthal , PC Repair and Maintenance- Fire wall Media, First Edition
R4	Manahar Lotia, Pradeep Niar, Modern Computer Hardware Course BPB Publication ,2011
R5	Andrew S. Tanenbaum ,David J. Wetherall Computer Networks – Prentice Hall fifth Edition
R6	William Stalling - Data Communication & Networks - Prentice Hall-Tenth Edition
R7	Fred Halsall -Data Communications, Computer Networks and Open Systems - Addison-Wesley, 1996
R5	Principles of Electronic Communications –Pradeep Kumar Ghosh

Online resources:

Sl.No.	Website Link
1	https://nptel.ac.in/courses/117101051/
2	https://www.tutorialspoint.com/digital_communication
3	https://nptel.ac.in/courses/108101113/
4	https://nptel.ac.in/courses/117/105/117105143/
5	https://www.slideshare.net/bhavyaw/noise-13468777